

Cartesian similarity is a poor proxy for transition-state quality: reference databases penalize valid alternative saddles

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Abstract

Machine-learning models for transition-state (TS) geometry prediction are almost universally benchmarked by the root-mean-square deviation (RMSD) of atomic positions, or by the mean absolute error of the interatomic-distance matrix (D-MAE), relative to a reference transition state. Both are Euclidean similarity measures. We show that they are poor proxies for the quantity that actually matters to a practitioner: whether a predicted geometry optimizes to a genuine transition state at a target level of density functional theory (DFT). Optimizing 520 interpolation-based guesses for 260 reactions from the Transition1x database at the r²SCAN-3c level, we find that *every* geometry-based quality measure - Kabsch-aligned RMSD, D-MAE, a reaction-coordinate projection, and cheap semi-empirical descriptors - saturates in a narrow band of the area under the receiver-operating-characteristic curve (AUC; defined in Section 2.4), 0.77–0.84, for predicting optimization success, with about 14% of low-RMSD guesses nonetheless failing. Decomposing the “failures”, 38% converge to a genuine but *different* first-order saddle point (median energy difference 1.36 eV from the reference, 77% of them *lower* in energy); these are legitimate alternative transition states that a reference-matching metric penalizes as errors. Only about 13% are symmetry-equivalent artifacts, which we remove algorithmically. Once RMSD is held fixed, no structural property of the reactant or product distinguishes converging from failing guesses: convergence is governed by the topology of the potential-energy surface (PES) - competing saddles and basins - which a Cartesian distance cannot see. A controlled experiment identifies the cause. Displacing each reference saddle by a *prescribed* distance along different directions - the reaction mode, the softest and stiffest vibrations, and isotropic noise - changes the probability of recovering it from 33% to 98% at identical distance, so distance alone cannot determine the outcome; yet once direction is held fixed as well, the ceiling persists at AUC $\approx$ 0.83. The saturation is therefore a limit on what a guess geometry can express, not a defect of any particular metric. All conclusions replicate on a second, chemically broader database covering second- and third-row elements, on which discrimination at matched guess quality is no better than on Transition1x, and on a state-of-the-art flow-matching generator: oracle

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best-of- N selection roughly triples its deployable single-draw success rate, yet no reference-free score recovers that gain, so reported best-of- N figures overstate what a user obtains by about threefold. We conclude that (i) the saturation near AUC 0.84 is an information ceiling rather than a shortcoming of any metric: a guess geometry simply does not encode how large the basin around its target saddle is, and the best geometric measure we construct already operates at that limit; (ii) RMSD and D-MAE compound this by measuring similarity to one arbitrarily chosen saddle rather than usefulness, while reference databases such as Transition1x are energetically skewed, so interpolation guesses routinely locate *lower-energy* saddles absent from the reference - whether these are alternative, stepwise, or unrelated transition states is, for these open-shell systems, beyond the reach of current automated structure perception (itself a cautionary result); and (iii) as a practical consequence, a cheap GFN2-xTB force descriptor evaluated on the guess predicts convergence about as well as oracle RMSD while requiring no reference structure - useful precisely because the ceiling means nothing better is available.

1 Introduction

Predicting transition-state (TS) geometries is a central problem in computational reaction chemistry, and a rapidly growing set of machine-learning models now addresses it — from the early graph-based generator TS-GCN [1] through diffusion models that build the saddle from a 2D graph [2] or from 3D reactant and product structures [3], to the recent optimal-transport and flow-matching approaches [4, 5, 6]. These are trained largely on the Transition1x database [7] and evaluated on held-out reactions from it. Almost without exception these models are evaluated by geometric similarity to a reference transition state - the root-mean-square deviation (RMSD) of atomic coordinates after optimal superposition, or the mean absolute error of the interatomic-distance matrix (D-MAE).

This is beginning to change, and our aim is to formalize the shift. A growing number of these models now report, alongside geometric error, the physically meaningful outcome of optimizing the predicted saddle at a quantum-chemical level: TSDiff [2], the machine-learning approach of Choi and co-workers [8], and the flow-matching model GoFlow [5] each quote a downstream saddle-optimization or recovery rate in addition to RMSD. Tellingly, TSDiff already observed that its samples include genuine transition states with barriers *lower* than the reference database [2] — scored there as deviations, but precisely the phenomenon we quantify in Section 3.2. What the field has not established is whether geometric similarity is merely an incomplete surrogate or a fundamentally *bounded* one, and what a downstream criterion must therefore measure. We address both.

The reason the field defaults to these Cartesian measures is practical rather than principled: the physically correct test of a predicted transition state - optimizing it to a first-order saddle and verifying, via an intrinsic reaction coordinate (IRC) [9], that it connects the intended reactant and product - requires Hessians and reaction-path integrations that are far too expensive to run for every prediction during model development. Machine-learned Hessians [10] and reactive interatomic potentials [11] lower this cost but do not remove the need for a target-level verification. A cheap geometric distance to a stored reference is an attractive surrogate. Implicit in its use is an assumption that, to our knowledge, has never been tested directly: that a lower similarity to the reference implies a more *useful* guess, i.e. one more likely to converge, under a standard saddle-point optimizer, to a valid transition state.

An analogous reckoning has already taken place in structure-based drug design, where similarity to a single crystallographic pose was long the standard benchmark. The PoseBusters analysis showed that deep-learning docking methods tuned to minimize pose RMSD routinely generate physically invalid geometries [12], and the CASF benchmarks established a distinction we return to

throughout: a model’s *sampling power* — placing a good candidate somewhere among its outputs — is separate from its *scoring power* — ranking that candidate first without access to the answer [13, 14]. Transition-state prediction, we will show, is subject to the identical split.

We test that assumption. We take deliberately simple, geometry-only predictors, optimize their guesses at a consistent density-functional-theory (DFT) level, and ask what - if anything - about a guess predicts whether it converges. The answer is sobering, and it points less at the models than at the way we benchmark them and at the reference data itself.

Our findings are the following. (1) Every geometric measure of guess quality we test saturates in a narrow band of predictive power, area under the receiver-operating-characteristic curve 0.77–0.84, and a purpose-built motion-weighted measure sits at the top of it. (2) A controlled experiment, in which reference saddles are displaced by a fixed distance along chosen normal modes, shows this is not a defect of the measures but an information ceiling: at identical distance the outcome swings from 33% to 98% with the *direction* of displacement, and even with direction fixed the ceiling persists — because reactions differ irreducibly in the size of the basin around their saddle, which no guess geometry encodes. (3) A large share of apparent failures are not failures but genuine, often lower-energy, alternative saddles that a reference-matching metric penalizes, exposing an energetic skew in the reference database. (4) The conclusions hold on a second, chemically broader database and, most consequentially, for a state-of-the-art generative model: oracle best-of- N selection roughly triples its deployable success rate, but no reference-free score recovers that gain, so reported best-of- N figures overstate what a user obtains. Throughout, we avoid drawing conclusions from quantities a practitioner cannot compute without already knowing the answer.

2 Methods

2.1 Data and reference

We use the Transition1x database [7] of reactant, product and transition-state geometries (originally at the ω B97x/6-31G(d) level). To obtain an internally consistent reference at our working level, we re-optimize each Transition1x transition state at the r²SCAN-3c composite DFT level [15] (ORCA 6.0.0 [16]). The set comprises 260 reactions (520 guesses); 259/260 reference saddles converge, all with exactly one imaginary vibrational frequency, confirming genuine first-order saddle points. Reactants and products are likewise optimized at r²SCAN-3c to define the minima used for connectivity. Areas under the curve (Tables 1 and 2, Figure 1) are reported on the 200-reaction batch (400 guesses) for which the complete descriptor set, including the GFN2-xTB features, was computed; the alternative-saddle census (Section 3.2) uses all 260 reactions (520 guesses).

To test generalization beyond first-row organic chemistry, we repeat the whole protocol on a second database, Reaction-QM [17]. We use only its density-functional-optimized subset (B3LYP-D3/TZVP), from which we sample 200 unimolecular reactions; these contain second- and third-row elements absent from Transition1x. All geometries are re-optimized at r²SCAN-3c exactly as above, so the two databases are compared at an identical working level and under an identical success criterion. Results are reported in Section 3.7.

2.2 Predictors

Two geometry-only predictors generate the guesses whose fate we study: *midpoint*, the linear interpolation $\frac{1}{2}(D^R + D^P)$ of the reactant and product distance matrices, embedded in three dimensions by classical multidimensional scaling (MDS); and *bond-order*, which models the length of each reactive bond from Pauling’s bond-order/bond-length relation and interpolates the remaining pairs.

These are not intended to be competitive with modern learned models; they are cheap, transparent sources of realistic guesses.

2.3 Downstream optimization and success

From each guess we run a transition-state optimization with analytic Hessians (ORCA `OptTS`) followed by a vibrational analysis. A guess is counted a *success* if the optimization converges, the result has exactly one imaginary frequency, and its Kabsch-aligned RMSD to the reference saddle is below 0.3 Å. Optimization cost is recorded as the number of geometry cycles and the wall-clock time.

2.4 Metrics and their evaluation

We compare guess-quality measures both requiring the reference TS (Kabsch RMSD; D-MAE; a change-weighted D-MAE; a projection of the aligned error onto the reaction coordinate, split into components parallel and perpendicular to it; the un-normalized root-square deviation; and a reactive-atom-only RMSD) and *deployable* descriptors requiring only the guess and the reactant/product (the GFN2-xTB force norm on the guess, and its energy relative to the endpoints). GFN2-xTB [18] denotes the second-generation geometry-, frequency- and non-covalent- interaction-parametrised semi-empirical tight-binding method.

Because the outcome is binary (success or failure), we quantify how well a continuous measure separates the two classes by the area under the receiver-operating-characteristic curve, abbreviated AUC. The receiver-operating-characteristic curve plots the true-positive against the false-positive rate as a decision threshold is swept; its area equals the probability that a randomly chosen failure is assigned a larger error than a randomly chosen success. An AUC of 0.5 is uninformative (a coin flip) and 1.0 is perfect separation; the measure is threshold-free and invariant to monotone rescaling.

2.5 Controlled displacement experiment

To test whether the saturation of the metrics is a property of the metrics or of the surface, we construct guesses whose deviation is fixed by design rather than observed. Each reference saddle is relaxed to a GFN2-xTB [18] saddle with the Sella optimizer [19], and the Hessian is computed there by finite differences. Displacement directions are taken as the imaginary mode, the softest and stiffest internal vibrations, and an isotropic Gaussian vector. Because Kabsch alignment removes rigid-body motion, the six translation/rotation modes are projected out of every direction before use — without this, a mode dominated by rotation produces a displacement an order of magnitude smaller than its nominal length. The step length is then solved by bisection so that the achieved Kabsch RMSD equals the target; the achieved value is stored for every trial and checked during analysis. The saddle search is repeated from each displaced structure and counted as a return if it converges within 0.3 Å of the original xTB saddle. This uses 187 (Transition1x) and 186 (Reaction-QM) reactions, about 3 200 trials per database, at GFN2-xTB throughout so that no reference-level inconsistency enters. Because the isotropic direction is drawn at random, its seed is derived from the reaction identifier rather than from the process, so that splitting the batch across cluster jobs cannot alter the result.

2.6 Generative model

To test whether the ceiling is specific to interpolation guesses, we also generate saddles with RitS [6], a flow-matching model that predicts a transition state from reactant and product structures. We use the released general checkpoint (not the Transition1x-specific one), which avoids train-test leakage on our Transition1x reactions; the reactant and product minima are the r²SCAN-3c geometries used everywhere else. Because the model is stochastic, we draw ten samples per reaction and keep them all. The predicted atom ordering matches the input, which we verify per reaction. We then optimize two of the ten samples at r²SCAN-3c under the same success criterion as the other predictors: the first draw (what a single call returns) and the sample closest to the reference saddle (the oracle best-of-ten). Reference-free selection among the ten is evaluated with three scores — the geometric medoid, the lowest GFN2-xTB force norm, and the lowest GFN2-xTB energy — none of which sees the reference.

2.7 Connectivity

For each converged *alternative* saddle we compute the intrinsic reaction coordinate (IRC) - the minimum-energy descent path in both directions - and identify the two endpoints by molecular-graph perception - covalent bond graphs and RMG-Py graph isomorphism, following the structure-perception approach of the Automated Rate Calculator (ARC) [20] - restricted where possible to the bonds that change across the reaction so that conformational differences are ignored. An endpoint is labeled reactant, product, or a distinct intermediate. Symmetry-equivalent artifacts (energy within 0.05 eV of the reference saddle) are removed before this analysis.

3 Results

3.1 No geometry metric predicts convergence beyond AUC ≈ 0.84

Two distinct binary outcomes are available, and we are explicit about which is meant throughout: *convergence*, whether the saddle search terminated at any first-order saddle at all, and *success*, the stricter requirement that it reached the reference saddle (within 0.3 Å, with exactly one imaginary frequency). All areas under the receiver-operating-characteristic curve quoted below refer to success unless stated otherwise; discrimination of mere convergence is uniformly weaker (RMSD: 0.733 against 0.806 for success), because a guess may converge perfectly well to the “wrong” saddle, which is the subject of Section 3.2.

Table 2 reports how well each measure predicts optimization failure ($n = 400$). Kabsch RMSD is the best single measure at AUC 0.806; D-MAE is markedly worse (0.769). Every measure - including the perpendicular reaction-coordinate error, which carries essentially all of RMSD’s signal while the parallel component (AUC 0.658) carries almost none - lands in a narrow band around 0.80. The distributions overlap heavily: about 14% of low-RMSD guesses fail and successes reach large RMSD, so no distance threshold cleanly separates the classes (Figure 2).

Two controls sharpen the point. The un-normalized root-square deviation ($\text{RMSD} \times \sqrt{N}$) gives AUC 0.808, statistically indistinguishable from RMSD: the $1/N$ size normalization carries no convergence information. A reactive-atom-only RMSD is *worse* (0.783): a hard binary restriction to the reacting center discards the whole-molecule fit that partly governs which basin the optimizer enters.

Between these extremes, a *continuous* weighting that concentrates on the reaction zone without discarding the rest improves on plain RMSD (Table 1). Weighting each atom’s squared deviation by its displacement across the reaction, $w_i = |\mathbf{r}_i^P - \mathbf{r}_i^R|$ (Kabsch-aligned), needs no free parameter

Weighting of the per-atom RMSD	AUC (failure)
uniform (plain Kabsch RMSD)	0.806
graph-distance decay from reactive atoms	0.819
Euclidean-distance decay from reactive atoms	0.820
displacement $ r^P - r^R $ (parameter-free)	0.839

Table 1: Continuous, reaction-zone-focused atom weights make RMSD a better convergence predictor than either the global or the reactive-atom-only form ($n = 400$).

and no explicit reaction center and reaches AUC 0.839; exponential decays in graph-bond distance and in Euclidean distance from the reactive atoms give 0.819 and 0.820. Thus the best purely geometric convergence measure is neither global RMSD nor a reactive-atom mask, but a motion-weighted RMSD — an improved, still oracle, benchmark metric.

Finally, a random forest [21] trained on purely deployable features (GFN2-xTB force and energy plus reactant/product geometry, with no reference TS) reaches AUC 0.791 - matching oracle RMSD without ever seeing the answer.

The limit is also not an artifact of the coordinate *system*. Repeating the analysis in redundant internal coordinates — the representation the optimizer itself uses — does not help: a reaction-change-weighted deviation over all bonds, angles and dihedrals (each expressed as an equivalent atomic displacement) reaches only AUC 0.59, because the many soft, periodic dihedral coordinates flood the metric; restricted to bonds and angles it recovers to 0.74, still below Cartesian RMSD. Weighting the coordinate types by the optimizer’s own default initial-Hessian stiffnesses (0.5, 0.2, 0.1 for stretches, bends and torsions) as a harmonic-energy metric does not help either (0.61): dihedrals are downweighted per coordinate but far outnumber the bonds, so their summed contribution still dominates. The result is unchanged when the metric is built on ORCA’s *exact* redundant internal coordinates, parsed from the optimization logs rather than reconstructed (0.60 against 0.59 for our bond-graph set). Added to the displacement-weighted RMSD, internal-coordinate deviations raise the cross-validated area under the curve from 0.839 to 0.844, within the fold-to-fold noise (± 0.02). Internal coordinates — even the optimizer’s own DOFs, weighting, and representation — thus carry no convergence information the Cartesian metric lacks.

Nor is the limit specific to *geometric* information. An electronic descriptor — the deviation of the guess’s Mayer bond orders from the reference, read from the same DFT output — reaches AUC 0.70 and, added to the displacement-weighted RMSD, again raises the cross-validated area under the curve only within noise (0.839 to 0.844). Reaction-level features computed from the reactant and product alone (their geometric separation, reaction energy, and total bond reorganization) predict convergence more weakly still (best single feature, the $R \rightarrow P$ displacement, AUC 0.70); the reaction energy is uninformative (0.46). These describe how demanding a reaction is on average, not whether a particular guess lies in a saddle basin, and so cannot exceed the guess-based ceiling.

Crucially, the limit is not an artifact of choosing a single metric. A random forest over all eight geometric measures (the five geometric rows of Table 2 and the three weighted variants of Table 1) reaches AUC 0.829, no higher than the single displacement-weighted RMSD, and adding the deployable xTB force does not improve on it either. Every guess-based measure we tried - geometric or semi-empirical, single or ensembled, oracle or deployable - lands in a narrow band that ceilings near AUC 0.84 (Figure 1). Convergence carries an irreducible unpredictability from any function of the guess, because the outcome is fixed by the potential-energy surface, not by the prediction.

We attribute the surprising transferability of a semi-empirical force to the r²SCAN-3c opti-

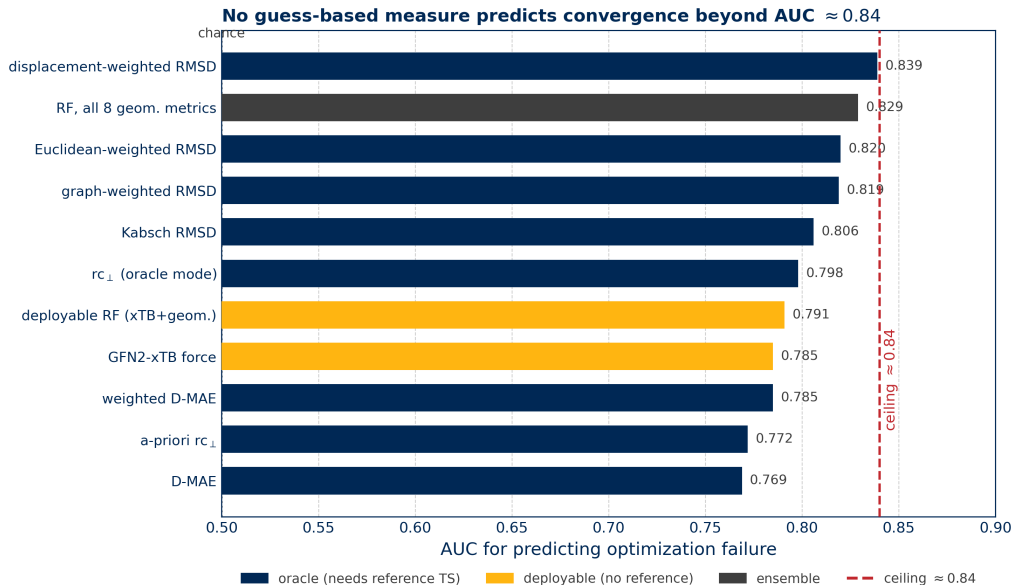


Figure 1: No guess-based measure predicts optimization convergence beyond AUC ≈ 0.84 ($n = 400$). Oracle measures (blue) require the reference transition state; deployable descriptors (green) require only the guess and the reactant/product; the random-forest ensemble (red) combines all eight geometric measures. None exceeds the single displacement-weighted RMSD.

mization outcome to the breadth of the transition-state basin: what governs convergence is not sub-Ångström geometric accuracy but whether the guess lies within the basin of attraction of the saddle, a region tens of degrees and tenths of an Ångström wide. GFN2-xTB, parametrised across the periodic table for structures and reaction energetics [18], reproduces this coarse curvature and gradient structure well even where it mis-estimates the barrier itself, so a small semi-empirical residual force is a faithful indicator that the higher-level optimizer will fall into the same basin. The force norm thus measures proximity to *a* stationary point on a correlated surface, which is exactly the quantity that predicts convergence.

3.2 A third of “failures” are alternative transition states

Of 239 failures, 142 (59%) do not converge, 92 (38%) converge to a *different* first-order saddle, and 5 collapse to a minimum. For the alternative saddles the median energy difference from the reference is 1.36 eV, and 77% are *lower* in energy; only 13% lie within 0.05 eV of the reference (symmetry-equivalent artifacts). A third of what the metric calls failure is therefore the optimizer locating a genuine, usually more stable, transition state that the reference database simply does not contain. This is consistent with reports from the generative-model literature itself: TSDiff, for instance, was observed to sample transition states with *lower* barrier heights than the reference database [2], which its RMSD-based evaluation scores as deviations rather than discoveries.

The magnitude matters kinetically. By transition-state theory [22] the rate constant is $k = (k_B T/h) \exp(-\Delta G^\ddagger/RT)$, so a median discrepancy of 1.36 eV (≈ 31 kcal mol⁻¹) in barrier height maps to a factor of $\exp(\Delta E/k_B T) \approx 10^{23}$ in the rate constant at 298 K. The estimate rests on two assumptions we state plainly. It equates the electronic barrier difference ΔE with a free-energy difference $\Delta G^\ddagger$, neglecting differential entropic and zero-point contributions; and, more importantly, it presumes the two saddles belong to the *same* reaction, so that the pre-exponential factor $k_B T/h$

Measure	AUC (failure)	Needs the reference TS?
Kabsch RMSD	0.806	yes
D-MAE	0.769	yes
Change-weighted D-MAE	0.785	yes
Reaction coord. $\perp$ error	0.798	yes
Reaction coord. $\parallel$ error	0.658	yes
GFN2-xTB force (guess)	0.785	no
Deployable random forest	0.791	no

Table 2: Every guess-quality measure saturates in the band 0.77–0.84. Deployable semi-empirical descriptors match oracle RMSD without a reference structure.

cancels [23]. Where an alternative saddle is instead a step of a *different* mechanism, the comparison is between rates of different reactions and the prefactors need not cancel. Even then, a barrier gap of this size dwarfs any plausible prefactor variation of a few orders of magnitude: a benchmark that treats these saddles as errors is not penalizing a small geometric slip but discarding transition states whose barriers differ by tens of orders of magnitude in rate — precisely the regime where getting the *right* saddle, rather than one close in RMSD, is decisive.

3.3 Convergence is a potential-energy-surface property

Restricting to low-RMSD guesses (< 0.28 Å), which “should” converge, and testing purely structural properties of the reactant and product, no feature separates the 30 failures from the 140 successes (all AUC 0.5–0.6). Reaction-amplitude features that appear predictive over all guesses (AUC 0.65–0.68) collapse to noise once RMSD is held fixed - they predict failure only by making the guess worse, not independently. The residual failures are structurally indistinguishable from the successes; their cause lies in the potential-energy surface (competing saddles, basins), not in the reaction’s gross structure.

3.4 Cost as a graded outcome

Convergence is binary, but its *cost* — the number of geometry-optimization cycles — is graded, and offers a second, more forgiving test of the metrics. Over the guesses that do converge, the geometry measures correlate with the cycle count only weakly to moderately (Spearman ρ : RMSD 0.59, perpendicular reaction-coordinate error 0.55, weighted D-MAE 0.54, D-MAE 0.51; the cheap GFN2-xTB force reaches 0.55). The same ordering as for success recurs, at a lower ceiling: a better guess is somewhat cheaper to optimize, but geometry explains well under half the rank variance in cost. A perfect guess (the reference TS itself) converges in ~ 5 cycles against ~ 20 for the interpolated guesses, so the potential cost saving is large — yet not predictable from Cartesian similarity.

3.5 Predictor comparison

The bond-order predictor marginally outperforms the midpoint baseline (success 56% vs 52%; 20.0 vs 21.3 optimization cycles, paired), but the effect is small, consistent with guess quality not being the limiting factor.

Metric distributions by outcome (n=400 guesses): heavy overlap — no RMSD cutoff separates usable from unusable guesses

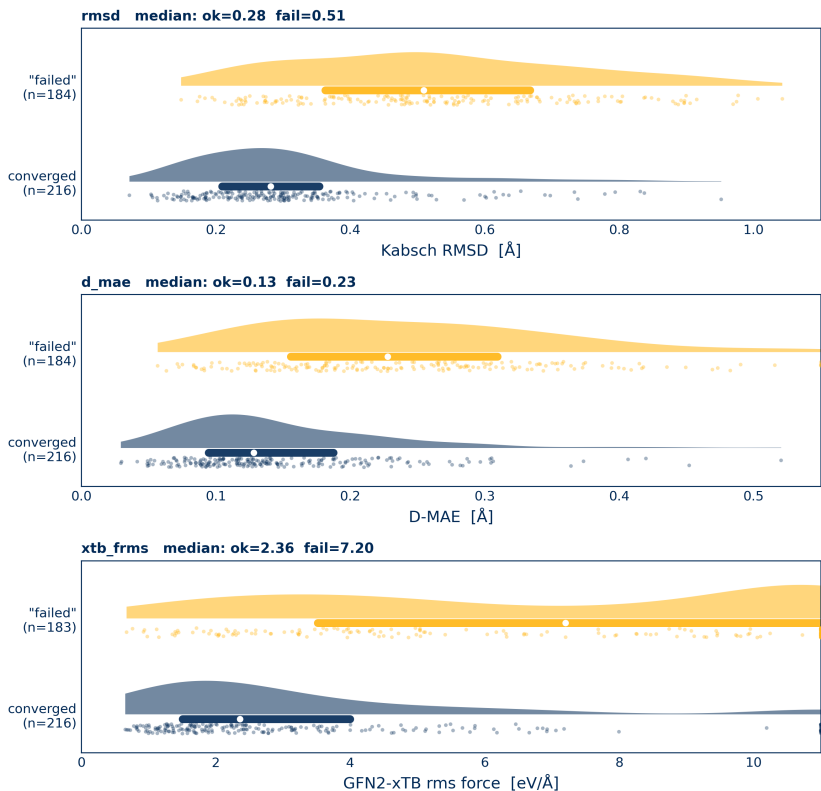


Figure 2: Distributions of each measure by outcome (raincloud: half-violin, box, and per-guess points). The heavy overlap, low-metric “failures” and high-metric successes, shows that no fixed RMSD cutoff separates usable from unusable guesses.

3.6 Why the ceiling exists: a controlled displacement experiment

The results so far are correlational: guesses differ from the reference saddle in both magnitude and character at once, so a metric’s failure could reflect a poor metric, a poor predictor, or an unforgiving surface. We separate these by constructing displacements whose distance is fixed by design. From each reference saddle — first relaxed to a self-consistent GFN2-xTB saddle so that the experiment is internally consistent — we step a prescribed Kabsch RMSD along four directions and repeat the saddle search: the imaginary reaction mode, the softest real vibration, the stiffest vibration, and an isotropic Gaussian direction. Rigid-body components are projected out and the step length is solved numerically, so every trial sits at the requested distance to better than 10^{-3} Å; the achieved distance is recorded for every trial and audited. A trial “returns” if the search converges to the same saddle within 0.3 Å.

At a fixed displacement of 0.20 Å, the return rate ranges from 33% along the reaction mode to 98% along the softest vibration on Transition1x, and from 22% to 96% on Reaction-QM (Figure 3); direction-independence is rejected overwhelmingly at every displacement in both databases (χ^2 on 3 degrees of freedom, $p < 10^{-16}$ throughout, reaching $p = 6 \times 10^{-36}$). The ordering is physically sensible and identical across the two databases: the reaction mode is the unstable direction and leaves the basin fastest, while the softest vibration is the most forgiving. This anisotropy is exactly what the topology of a saddle region predicts — reaction paths curve, branch, and pass through

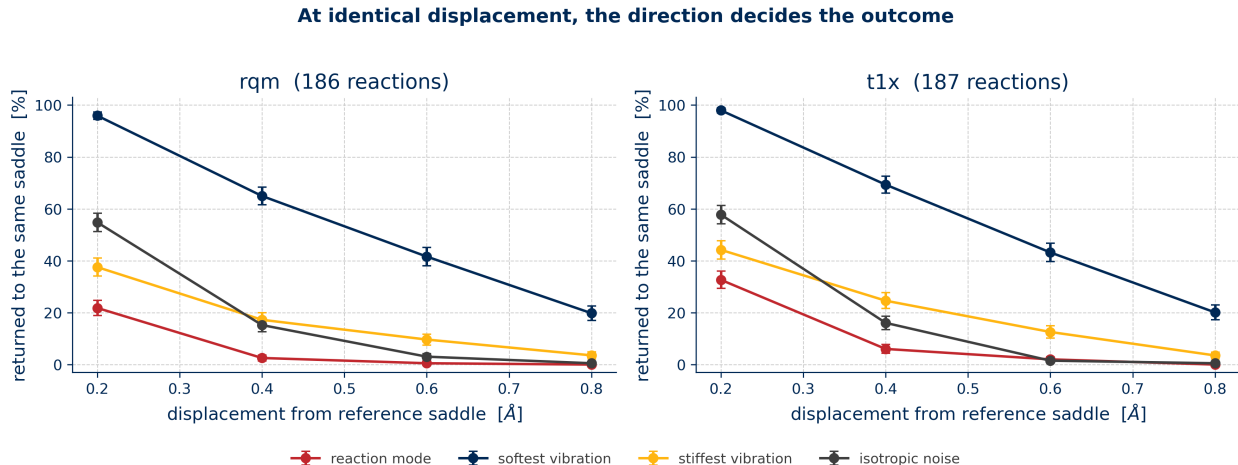


Figure 3: Controlled displacement from the reference saddle (187 and 186 reactions, $\sim 3\,200$ trials per database). Every point at a given abscissa sits at an *identical* distance from the saddle — verified to 10^{-3} Å for every trial — so the vertical spread is due to direction alone. Error bars are binomial standard errors.

valley-ridge inflections [24], so displacement along the unstable eigenvector escapes the basin far sooner than an equal step along a soft orthogonal mode. *Cartesian distance therefore cannot determine the outcome, because outcomes spanning 33% to 98% share the same distance.*

Controlling direction as well recovers part, but only part, of the lost discrimination. Pooled over directions, the area under the curve for displacement predicting failure to return is 0.770 (Transition1x) and 0.757 (Reaction-QM); restricted to a single direction it rises to 0.828 and 0.822. Direction-blindness thus costs roughly 0.06 in area under the curve — a real but modest penalty, and far from the whole gap to a perfect predictor. The residue is the point: even with the distance controlled to 10^{-3} Å, the direction held fixed, and the reference saddle identical, displacement predicts return only at $\text{AUC} \approx 0.83$. That is essentially the value reached by the best purely geometric measure on real predictions (displacement-weighted RMSD, 0.839, Table 1).

Because these displacements are drawn from a small set of prescribed lengths, almost every pair of trials is tied in distance; areas under the curve are therefore computed with mid-ranks, without which ties are broken arbitrarily and the statistic becomes meaningless.

The experiment was repeated at the working level of theory to confirm that it is not an artifact of the semi-empirical surface: 320 additional $r^2\text{SCAN-3c}$ saddle optimizations on 40 Transition1x reactions, with the displacement directions taken from the reference DFT Hessians rather than recomputed. The direction effect reproduces (98% return along the softest vibration against 28% along the reaction mode at 0.20 Å; $p = 3 \times 10^{-8}$ and 8×10^{-14} at the two displacements). Compared like for like — the same 40 reactions and the same two displacements — the two levels of theory agree to within 0.01 in area under the curve (pooled 0.651 at $r^2\text{SCAN-3c}$ against 0.665 at GFN2-xTB; within-direction 0.711 against 0.722), so the semi-empirical surface is a faithful proxy here and the larger xTB sample can carry the statistical weight. The lower absolute values in this comparison are themselves an instance of the range effect of Section 3.7: restricting the same xTB data from four displacements to two moves the within-direction area from 0.828 to 0.702.

Two independent routes therefore arrive at the same number. This is the mechanism. The ceiling is not an artifact of metric design, nor merely of RMSD’s blindness to direction, nor of noise in the predictors: it survives an experiment in which all three are eliminated or controlled by

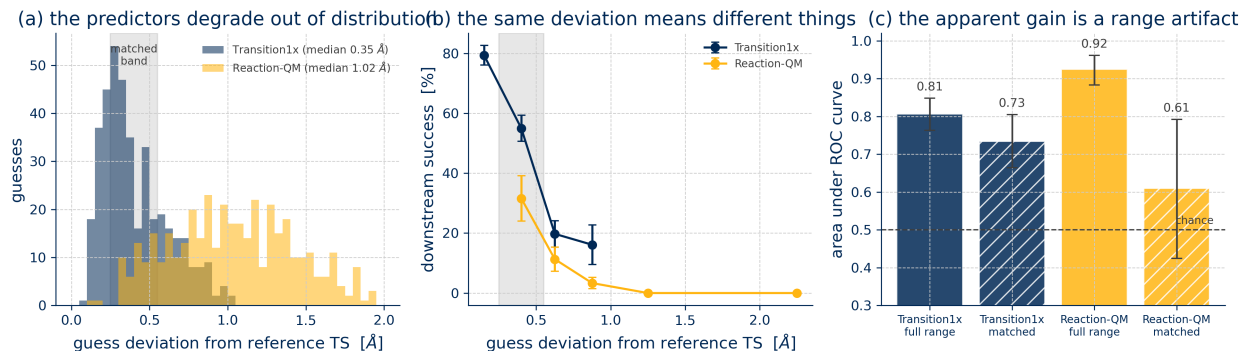


Figure 4: Generalization to Reaction-QM. (a) The predictors degrade sharply out of distribution; the shaded band is the Transition1x interquartile range of guess deviation, used to match the two datasets. (b) Downstream success against guess deviation: at every matched deviation Reaction-QM succeeds less often, so a threshold calibrated on one database does not transfer to the other. (c) The apparent improvement in discrimination on Reaction-QM (0.92) is an artifact of its much wider deviation range; restricted to the matched band it falls to 0.61, statistically indistinguishable from both the Transition1x value and from chance. Error bars are binomial standard errors in (b) and 95% bootstrap confidence intervals in (c).

construction. What remains is irreducible variation between reactions in how large a basin their saddle possesses — a property of the potential energy surface that no measurement of the guess geometry can access. Metrics saturate near 0.83 because that is where the information in a guess geometry runs out, and the best of them already operates at that limit.

A corollary bears on how synthetic guesses should be built. Isotropic Gaussian noise, the natural way to manufacture guesses of a prescribed quality, does not behave like a typical direction: it tracks the *reaction mode* arm, the harshest of the four. Perturbing a reference structure isotropically therefore understates the success rate a real predictor would achieve at the same nominal deviation, and is not a neutral surrogate for predictor error.

3.7 Generalization to a second, harder database

All results above derive from a single database of small first-row organic reactions. To test whether they are an artifact of that chemistry, we repeated the entire protocol on 200 unimolecular reactions drawn from the density-functional subset of Reaction-QM [17], which extends to second- and third-row elements (Si, P, S, Cl) and larger systems. Reference saddles behave as before (197/200 converged, all with exactly one imaginary frequency), so the comparison is not confounded by a poor reference.

Three things change, and one does not (Figure 4). First, the predictors degrade sharply out of distribution: the median guess deviation rises from 0.35 Å to 1.02 Å, and only 2 of 400 guesses fall below 0.3 Å (against 155 of 400 on Transition1x). Second, downstream success collapses from 54% to 6%, with convergence falling from 73% to 34% and the median cost of a converged optimization rising from ~20 to ~32 cycles. Third, the fraction of converged guesses that reach an alternative saddle rather than the reference one *grows*, from 19% to 26%: as reactions become more complex, the reference transition state accounts for a smaller share of the accessible saddles, reinforcing Section 3.9.

What does not change is the central negative result. Taken over the full range, the area under the receiver-operating-characteristic curve for the deviation measure appears to *improve* to 0.92,

which would suggest the ceiling of Section 3.1 had been broken. It has not. That figure is an artifact of range: discrimination is trivially easier when the guesses span a wide quality range (0.74–1.33 Å interquartile) than a narrow one (0.25–0.54 Å). Restricting the Reaction-QM guesses to the Transition1x interquartile range removes the advantage entirely — the area under the curve falls to 0.61 (95% bootstrap confidence interval 0.42–0.80, $n = 53$), statistically indistinguishable from both the Transition1x value of 0.73 (0.66–0.81) and from chance. Geometric similarity to the reference transition state is therefore no more predictive on a harder and chemically broader database than on the one the conclusion was drawn from; if anything the mapping from deviation to outcome is itself database-dependent, so a threshold calibrated on one dataset does not transfer to another.

The one result that does replicate is the predictor ordering: the bond-order predictor again edges out the midpoint baseline (7% vs 5% success), the same direction and the same negligible magnitude as before.

3.8 The same ceiling constrains a state-of-the-art generative model

The predictors studied so far are deliberately simple. A natural objection is that the ceiling is a property of weak, interpolation-based guesses and would lift for a modern machine-learned generator. We therefore repeated the analysis with RitS [6], a flow-matching transition-state generator that takes the reactant and product geometries and emits a predicted saddle. We use the authors’ released checkpoint, which is not trained on Transition1x, so our evaluation carries no train–test leakage. Two limits of the tool bound its coverage and are reported rather than hidden: it supports only the elements C, H, N, O, S, F, P (removing 96 of the 200 Reaction-QM reactions), and its bond perception from Cartesian input fails outright on 16 of 200 Transition1x and 7 of the remaining 104 Reaction-QM reactions, leaving 184 and 97 reactions.

The first observation is that the generator is *stochastic*: each call draws an independent sample, and the ten draws for a single reaction span a median range of 0.67 Å on Transition1x (Figure 5a) — wider than the entire spread between the methods compared in the rest of this paper. This makes the choice of what to report decisive. The single draw a user actually obtains has a median deviation of 0.82 Å (Transition1x) and 1.08 Å (Reaction-QM), no better than the linear-interpolation baseline. The *best* of ten draws, selected by proximity to the reference saddle, has a median deviation of 0.37 and 0.73 Å — a two- to threefold improvement, and the number a best-of- N benchmark would report.

That best-of- N number is an oracle. Selecting the closest sample requires the reference transition state, which is exactly the unknown the model exists to predict. The gap it opens is not academic: optimized at r²SCAN-3c under our success criterion, the deployable single draw reaches the reference saddle for 16% (Transition1x) and 4% (Reaction-QM) of reactions, while the oracle best-of-ten reaches it for 44% and 23% (Figure 5c; McNemar $p < 10^{-3}$ on both, discordant pairs 54 vs 2 and 18 vs 0). Oracle selection roughly triples the success rate — a gain entirely unavailable at prediction time.

The obvious question is whether a *reference-free* score can recover it. We tried three: the consensus (medoid) sample, which requires no electronic structure; the lowest GFN2-xTB force, which is the deployable descriptor this paper recommends elsewhere; and the lowest GFN2-xTB energy. None works. Across both databases the sample each selects sits at a mean rank of 4.5–5.7 among the ten (Figure 5b), against a chance expectation of 5.5 and the oracle’s 1.0; the consensus sample is marginally better than chance and the energy and force selectors are not. The near-threefold best-of- N advantage is therefore real but structurally inaccessible: it lives in knowing which sample is closest to an answer one does not have, and no cheap proxy stands in for that

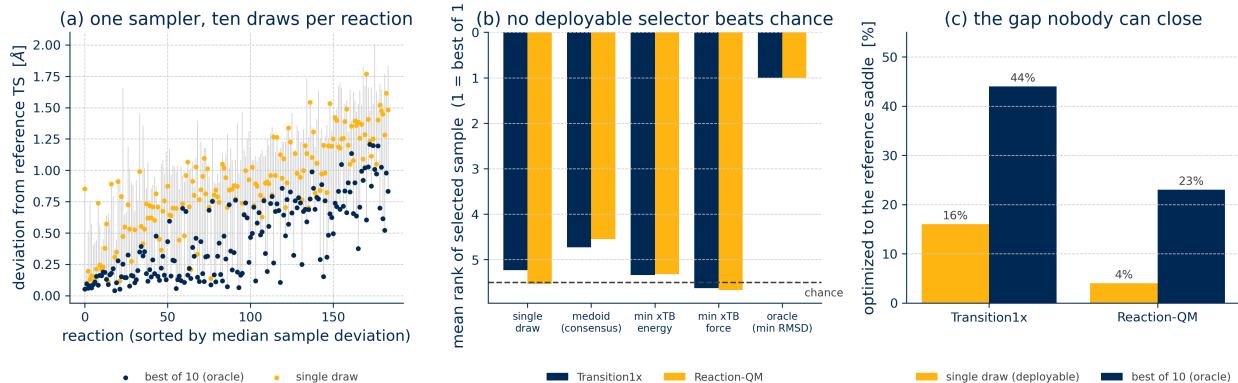


Figure 5: A state-of-the-art generator under the same lens. (a) RitS is stochastic: for each Transition1x reaction (sorted by median), ten draws span a wide range; the oracle best (blue) sits far below the single draw (orange). (b) Mean rank of the sample chosen by each reference-free selector, out of ten (1 = best); every deployable selector lands near the chance line, only the oracle (min RMSD to the reference) reaches 1. (c) Optimized at r²SCAN-3c, the oracle best-of-ten roughly triples the deployable single-draw success on both databases — a gap no reference-free selector closes.

knowledge. The information ceiling of Section 3.1 thus reappears for a modern generator in a sharper form — as a selection problem — and it cautions that best-of- N scores in the generative-TS literature overstate deployable performance by roughly a factor of three.

3.9 The nature of the alternative saddles is not reliably determinable

We attempted to classify each alternative saddle by computing its intrinsic reaction coordinate (IRC) at r²SCAN-3c and identifying the two endpoints (after a short GFN2-xTB relaxation of the frequently truncated IRC endpoints). Reference-transition-state controls, whose IRCs must connect their reactant and product, validated the endpoint extraction (6/6). Symmetry-equivalent artifacts (energy within 0.05 eV of the reference) account for $\sim 13\%$ of the alternative saddles and were removed.

However, assigning the remaining endpoints to a chemical species proved *not robust*. Four reasonable identity criteria - covalent adjacency graph, InChI, RMG-Py graph isomorphism (as used by the Automated Rate Calculator, ARC [20]), and canonical SMILES - returned between 6% and 91% “same reaction” on the identical set of endpoints. Manual inspection identified three specific algorithmic blind spots, all aggravated by the open-shell character of many Transition1x species. First, *radical-center and multiplicity assignment* is not consistent across two geometries of the same molecule: perception routines placed unpaired electrons on different atoms and returned different multiplicities for a species and its own relaxed copy. Second, *resonance-structure selection* is arbitrary - a delocalised radical is represented by one of several equivalent Kekulé/resonance forms, and graph isomorphism then declares two forms of the same species distinct. Third, *path directionality* is lost: an IRC endpoint may relax to a conformer or tautomer of the reactant/product that the reference minimum does not match, and the two IRC branches occasionally invert, so reactant and product labels swap. The net effect is that a single species is repeatedly split into spurious “intermediates” (many carried a SMILES string identical to the product yet were flagged distinct; Figure 6). We therefore *cannot* state with confidence whether the alternative saddles are concerted alternatives of the same reaction, genuine stepwise elementary steps, or saddles of different reac-

tions. This is itself a finding: for open-shell reaction networks, even the seemingly simple question “did the guess find the intended reaction?” is not answerable by current automated structure perception, and a robust, perception-independent connectivity criterion is an open problem. What we can state solidly, the foregoing notwithstanding, rests on energetics alone: the alternative saddles exist, are usually lower in energy, and are penalized by a reference-matching metric.

4 Discussion

Cartesian distance is a flawed proxy for surface topology. RMSD and D-MAE measure proximity to *one* recorded saddle, whereas what a practitioner needs is that the guess lie in the basin of attraction of *a* valid saddle connecting reactant and product - a topological property. The two notions coincide only when the reference is the unique relevant saddle, which we find fails roughly a third of the time.

Reference databases are energetically skewed. Interpolation guesses routinely optimize to lower-energy saddles (77% of the alternative-saddle cases) absent from Transition1x. A geometrically “wrong” prediction frequently reaches a more stable transition state. Benchmarking against a single reference rewards a model for reproducing the database’s particular saddle choice rather than for locating a valid, possibly better, transition state - even though (Section 3.9) we cannot robustly determine, for these open-shell endpoints, whether each lower saddle belongs to the same reaction, a stepwise variant, or a different one. That a single stored saddle is an arbitrary choice rather than ground truth is corroborated by reaction databases built to enumerate mechanistic multiplicity: RGD1, for example, catalogs many reactions admitting more than one validated transition state [25].

Local optimizers are Hessian-blind. A quasi-Newton saddle-point optimizer only ever sees the local Hessian; from a given guess it walks to whichever saddle the local curvature indicates, which need not be the closest in RMSD nor the reference. This is why no function of the guess geometry predicts convergence beyond $\text{AUC} \approx 0.84$: the outcome is fixed by the surface, not the guess. Our controlled-displacement experiment makes this concrete — direction at fixed distance swings the outcome from 33% to 98%, and even the direction-controlled ceiling matches the best real metric — so the saturation is an information limit, not a modeling artifact.

For generative models the ceiling becomes a selection problem. A stochastic generator can place a good saddle among its samples far more often than any single draw suggests: for RitS the oracle best of ten roughly triples the deployable success rate. But choosing that sample requires the reference, and none of the reference-free scores we tried — consensus, xTB force, xTB energy — selects better than chance. This is the sampling-power/scoring-power split that structure-based drug design formalized a decade ago [13, 14]: generating a good candidate somewhere in the top- N is a different, easier problem than ranking it first without an oracle, and the gap between them is a general bottleneck of generative models, not a quirk of RitS. Best-of- N evaluation therefore measures the wrong quantity: it credits a model for a sample it cannot identify without the answer, and overstates deployable performance by about threefold. The honest figure is the expected quality of a *single* draw, or of a draw selected by a genuinely reference-free rule.

Toward better objectives. The community should (i) report convergence rate and IRC connectivity rather than RMSD/D-MAE; (ii) score against *any* valid reactant→product saddle rather

than a single reference; (iii) for stochastic generators, report single-draw or reference-free-selected quality rather than oracle best-of- N ; and (iv) explore topological or energy-derivative-free training objectives (for instance a force-norm or saddle-character penalty) that reward residence in a saddle basin over proximity to a particular geometry. For a cheap convergence proxy at scale, a GFN2-xTB force descriptor on the guess is about as informative as oracle RMSD and needs no reference structure.

5 Conclusion

Geometric similarity to a reference transition state does not predict whether a predicted geometry is useful: every geometry metric saturates in a narrow band below AUC 0.84, and a large share of apparent failures are lower-energy saddles the reference database omits and the metric penalizes. A controlled experiment locates the cause. At a fixed distance from the saddle the outcome varies from 33% to 98% depending only on the direction of displacement, so distance cannot determine convergence; yet once direction is *also* controlled the ceiling persists at AUC ≈ 0.83 , and the effect reproduces at r²SCAN-3c. The saturation is therefore not a deficiency of any particular metric but a limit on what guess geometry can express: reactions differ irreducibly in the size of the basin around their saddle. This holds on a second, chemically broader database: once the comparison is matched for guess quality, geometric similarity is no more predictive there than on the database the conclusion was drawn from, and the mapping from deviation to outcome does not transfer between the two. The same limit constrains a state-of-the-art generative model: its oracle best-of- N score, roughly triple its deployable single-draw success, cannot be recovered by any reference-free selection rule, so reported best-of- N figures do not reflect what a user obtains. Transition-state prediction should be benchmarked by surface behavior - convergence, and where determinable, connectivity - not by Cartesian distance, and generative models by the quality of a single or reference-free-selected draw rather than an oracle best-of- N . That determining connectivity is itself hard for open-shell reaction networks only underscores that geometric distance is the wrong yardstick.

Data and code

Analysis scripts (`bench/`) and reduced datasets (`server/`) are provided; DFT jobs were generated by `make_orca_jobs.py`, `make_orca_ladder.py` (controlled displacement) and `make_orca_rits.py` (generative guesses), and reduced on the cluster to compact JSON by `reduce_batch.py` / `reduce_ladder.py`. RitS guesses and reference-free selectors are produced by `run_rits.py` and `rits_selectors.py`.

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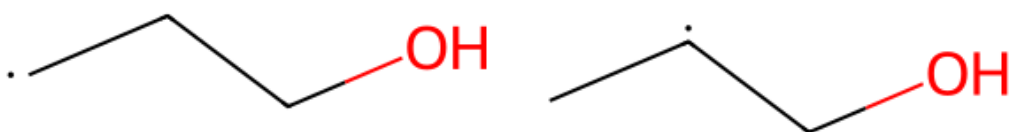
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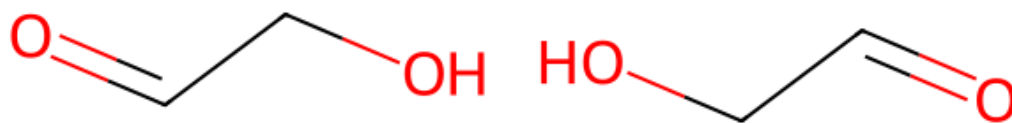
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Resonance ambiguity



Radical-site migration



Path-direction inversion

Figure 6: Three algorithmic blind spots that make automated species perception inconsistent for open-shell IRC endpoints, each illustrated by a pair that is chemically one species yet is repeatedly assigned two identities. *Left/right within each row*: (top) resonance ambiguity, a delocalised radical represented by two equivalent forms; (middle) radical-site migration, the unpaired electron placed on a different atom for two geometries of the same molecule; (bottom) path-direction inversion, an IRC whose two branches relax to reactant/product labels that swap. These failures underlie the 6%–91% spread across identity criteria in Section 3.9.